Problem 02: Prompt Pit

Time limit: 1.5 seconds

Two prompt-engineers face off in the Prompt Pit, a pick-ban arena where they assemble teams of AI models from a shared model pool. Each model has a known score, and once a model is chosen or banned, it disappears from the pool forever – thrown into the deprecated weights folder.

The draft follows a fixed sequence of actions. On each turn, the active prompt engineer must perform exactly one action:

1. Pick a model
2. Ban a model

If an engineer misses a pick, the system automatically assigns them a random model. If they miss a ban, nothing happens.

Each engineer tries to maximize the final score difference in their favour.

Given all model strengths and the full pick/ban sequence, determine which engineer (**A** or **B**) has the advantage at the end of the draft, and by how much.

It is guaranteed that there are no ties.

Input:

The first line contains a single integer **N (2<=N<=100)** – the number of models in the model pool.

The second line contains **N** integers $a_1, a_2, a_3, \ldots, a_N$ $(1 \le a_i \le 10^8)$ – the score of all the models.

The third line contains a single integer M $(2 \le M \le min(N, 20))$ – the number of actions the prompt engineers must perform.

Next M lines look like **"ENGINEER ACTION"**, where engineer is denoted by the letters **A** or **B**, and action is either a **"pick"** or **"ban"**

It is guaranteed that both engineers make at least **1** pick, and both have the same number of picks and same number of bans

Output:

Print the engineer (**A** or **B**) which wins and the absolute difference in score if both perform actions optimally.

Sample Input 1	**Sample Output 1**
5 3 7 2 9 5 4 A pick B pick A ban B ban	A 2

Sample Input 2	Sample Output 2
6 10 1 9 8 3 2 6 A ban B ban A pick B pick A pick B pick	A 3